EEHC DISTRIBUTION MATERIAL SPECIFICATION	EDMS 11-400-1
	Date: 3-08-2021

EDMS 11-400-1

SPECIFICATION FOR DISTRIBUTION SURGE ARRESTER
(INDOOR – OUTDOOR)
12 & 24 kV

Issue: Aug-2021 / Rev- 1

EDMS 11-400-1

SPECIFICATION FOR DISTRIBUTION SURGE ARRESTER

(INDOOR – OUTDOOR)

12 & 24 kV

Issue: Aug-2021 / Rev- 1

CONTENTS

1. Definition:

A surge arrester is a protective device for limiting surge voltages on equipment by diverting surge current and returning the device to its original status. A surge arrester is capable of repeating these functions a large number of times as specified.

2. SCOPE:

This specification describes the minimum technical requirements for design, engineering, manufacturing, testing, inspection and performance of surge arresters intended to be used on 12 kV and 24 kV medium-voltage distribution network .(Indoor and outdoor)

The surge arresters shall be Metal Oxide (MO) without spark gap with silicon or porcelain housing.

3. APPLICABLE STANDARDS

Unless otherwise specified in this specification should be comply with the latest edition of IEC standard and should be designed, manufactured and tested in accordance with the latest applicable IEC standards as follows: **Table (1)**

Table (1)

Standard	Description
IEC 60099-4	Surge arresters – Part 4: Metal-oxide surge arresters without gaps for A.C. systems
IEC 60099-5	Surge arresters – Part 5: Selection and application recommendations
IEC 60099-6	Surge arresters – Part 6: Surge arresters containing both series and parallel gapped structures – Rated 52 kV and less
IEC 60099-8	Surge arresters – Part 8: Metal-oxide surge arresters with external series gap (EGLA) for overhead transmission and distribution lines of A.C. systems above 1 kV
IEC 60071-1	Insulation coordination – Part 1: Definitions, principles and rules

IEC 60071-2	Insulation coordination – Part 2: Application guide
IEC 60060-1	High-voltage test techniques. Part 1: General definitions and test requirements
IEC 60507	Artificial pollution tests on high-voltage ceramic and glass insulators to be used on A.C. systems
IEC 60815-1	Selection and dimensioning of high-voltage insulators intended for use in polluted conditions – Part 1: Definitions, information and general principles
IEC 60815-2	Selection and dimensioning of high-voltage insulators intended for use in polluted conditions – Part 2: Ceramic and glass insulators for A.C. systems
IEC 60815-3	Selection and dimensioning of high-voltage insulators intended for use in polluted conditions – Part 3: Polymer insulators for A.C. systems
IEC 60038	IEC standard voltages
IEC 62217	Polymeric HV Insulators for Indoor and Outdoor Use – General Definitions, Test Methods and Acceptance Criteria
IEC 60060	High-Voltage Test Techniques
IEEE C62.11	IEEE Standard for Metal-Oxide Surge Arresters for AC Power Circuits (> 1 kV)
IEEE C62-22	IEEE Guide for the Application of Metal-Oxide Surge Arresters for Alternating Current systems

4. ENVIRONMENTAL CONDITIONS

The performance of the surge arrester should be guaranteed for the following environmental conditions, any differences in the guaranteed performance should be clearly set out in the offer:

Table (2)

Minimum ambient temperature	-5°C
Maximum ambient temperature	45°C (50 °C as option).
Maximum relative humidity	95%.
Maximum altitude	1000 m.

5. DESIGN CRITERIA

i. The surge arresters shall be gapless Metal Oxide (MO) non-linear resistor type and is housed on a sealed casing to prevent ingress of moisture and dust.

ii. The surge arresters shall be distribution heavy- duty class with 10kA nominal discharge current for both 12 kV and 24 kV

iii. The housing of the arrester should be designed for maximum withstand of lightning and switching over voltage impulses. The housing shall be free flaws, voids, conducting impurities cracks and chipped edges.

iv. The surge arrester shall be well designed and constructed to have :

a) Maximum withstand of temporary over voltages (TOV)

b) Thermal stability (no thermal run a way) i.e. maximum thermal energy absorption capacity and maximum thermal dissipation

c) Maximum insulation safety margin i.e minimum lightning and switching protection levels.

d) Maximum protective zone (length).

e) Maximum withstand of all current impulses which it can be subjected to during operation.

v. The surge arrester shall be dust and moisture proof. It shall be hermetically sealed to assure Positive proofing throughout its life according to IEC 60099-4.

vi. The arresters line - lead terminal shall accommodate all normal sizes of copper or Aluminum Conductors with diameter from 5 to 20 mm with length 75 cm .

vii. The arrester shall be supplied with all necessary clamps , straps , and fittings for the purpose of its mounting on steel lattice pole . Also, each arrester shall be supplied with its relevant arrester disconnector.

6. Technical specifications for the surge arrestor

Unless otherwise specified, the surge arresters shall be designed to operate on the system parameters mentioned in **Table 3**

Table (3)

SYSTEM PARAMETER	SYSTEM	
NOMINAL VOLTAGE	11KV	22 KV
Highest system voltage	12 kV	24 kV
Continuous Operating Voltage (Uc)[1]	≥ 9.6	≥ 19.2
Neutral Arrangement	Effectively Grounded & Non-Effectively Grounded	
Rated Frequency	50 Hz	
Lightning protection level ,(at current impulse 8/20µs, 10 kA), not more than	35 kVpeak	70 kVpeak
Switching protection level , (at current impulse 30/60µs, 250 A), not more than	28 kVpeak	56 kVpeak
long duration current impulse duration of lightning impulse residual voltage before and after test	≤ 5%	
Short-Circuit withstand current for duration 0.2sec, KA rms not less than	20 kA	
High current , impulse 4/10µs, kA peak not less than	100 kA peak	
Creepage Distance	2.5 cm/kV (Indoor) 4 cm/kV (Outdoor)	
Nominal Discharge Current Based on 8/20 µs	10 kA	
Steep front current impulse (10 kA) residual kV peak not more than	40 kVpeak	80 kVpeak
one – minute power frequency wet withstand voltage insulating housing, not less than	30 kVr.m.s	60 kVr.m.s
Impulse 1.2 /50 µs withstand voltage of insulating housing not less than	46 kVpeak	92 kVpeak
Energy absorption capability , kJ / kV of rated voltage (Ur) not less than	2.8 kJ / kV	

7. INSPECTION AND TESTING

7.1. Routine Tests

Routine tests in conformance with the applicable clauses of IEC 60099-4 or equivalent shall be performed on all surge arresters. Additional tests such as acceptance tests and special thermal stability test in compliance with the applicable clauses of IEC 60099-4 : -

- Measurement of reference voltage (Uref).
- residual voltage tests .
- partial discharge test.

7.2. Type tests

Type test should be performed in complete conformance with the applicable clauses of IEC 60099-4 or equivalent. It shall be performed at accredited laboratories.

A representative from the Distribution Company could attend these tests before acceptance.

- Insulation withstand test
- Residual voltage tests
- long duration current impulse with stand test.
- operating duty tests.
- tests of arresters disconnectors .
- Artificial pollution test (CERAMIC &GLASS) .
- Partial discharge test.
- Lighting impulse residual voltage test.

7.3. complete arrester

- Visual inspection.
- Measurement of creepage distance.
- Measurement of power frequency voltage.
- Partial discharge test.

7.4. individual arrester housing:

- Power frequency wet withstand voltage test.
- 1.2 /50 µs lighting impulse withstand voltage test.
- Porosity test (in case of porcelain housing) according to latest IEC.

8. Marking

- Each insulator will be clearly and indelibility marked with the Following
- Manufacture's name and country
- Year of manufacture
- Nominal current
- Maximum Continuous Operating Voltage (Uc) or (M.C.O.V)
- Rated voltage
- Absorption capacity kJ / kV.

9. Packing

Packing and delivery shall generally be in suitable (wooden crates for porcelain and harden cartoon for polymers) and in such a manner so as to prevent damage to them.

- Each box shall bear the following information legibly printed in a visible place:
- Purchase order number / Tender number
- Manufacturer's name
- Year of manufacture
- Quantity of pcs. in the box
- Gross weight in kilograms (kg)
- Position of slinging points and other relevant handling instructions

10. Catalogues And Samples

The necessary technical catalogues containing the drawing and full technical. Date shall be submitted with the offer The tenderer shall submit (1) arrestor of each type free of charge for the purpose of offer analysis. The quality control procedure shall be submitted with the offer .

EEHC DISTRIBUTION MATERIAL SPECIFICATION	EDMS 11-400-1
	Date: 3-08-2021

GUARANTEE TABLE

`No.	Item	Spec.
1.	Type and manufacturer	
2.	Nominal operating voltage	
3.	Maximum continues operating voltage (Uc) or (MCOV) , kV r.m.s	
4.	Degree of protection	
5.	Material of housing	
6.	Dimensions : * Height : cm * Diameter : cm * Weight : Kg	
7.	Nominal discharge current , impulse 8/20 µs , kA peak	
8.	Lightning impulse Protection level , kV peak , at current impulse 8/20 µs at 10 kA	
9.	Lightning impulse residual voltage , K A peak, at current impulse 8/20 µs at 10 kA	
10.	switching impulse residual voltage , kV peak, at current impulse 30/60 µs	
11.	Long duration impulse 2000µs, kA peak	
12.	High current impulse 4/10µs, kA peak	
13.	Residual voltage , KV peak, at steep front current impulse at 10 kA	
14.	Energy absorption capability kJ/kV of rated voltage	
15.	Short circuit withstand current for duration 0.2 s, kA r.m.s	
16.	Creepage distance of insulating housing, mm * Indoor : * Outdoor :	
17.	One minute power frequency withstand voltage of insulating housing, kV r.m.s	
18.	Iimpulse 1.2/50 µs withstand voltage of insulating housing, kV r.m.s	
19.	Insulation co-ordination safety margin % (at security factor=1.2)	
20.	Protective zone (length),m (at security factor=1.2&sreepness of U kV/µS)	
21.	Temporary over voltage (TOV), V r.m.s - continuous : - 10 s : - 1s :	

We guarantee the data given above for the equipment offered.

Signature: ……………………………… Date: ………………………………